Model Limits and Decision Context

ECON 417 Business Forecasting

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How Lecture 5 Fits the Course

- Lecture 4 gave us richer point forecasts.
- Lecture 5 asks: **how wide is the range of plausible outcomes?**
- Lecture 6 will turn those uncertainty measures into monitoring and governance rules.

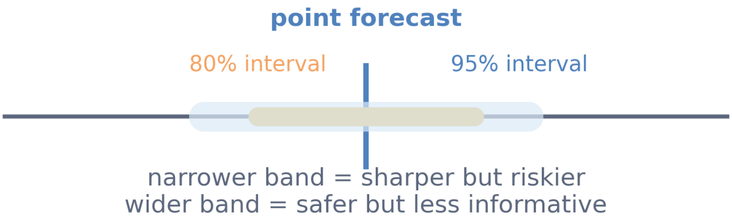
Key message: A point forecast without uncertainty can create false confidence in risk decisions.

Agenda (Today)

1. Point Forecast vs Interval Forecast
2. Building and Interpreting Intervals
3. State-Dependent Uncertainty
4. Scenario Forecasting
5. Asymmetric Costs of Forecast Errors
6. Decision Interpretation

Point Forecast Is Not the Whole Story

A point forecast is not a complete forecast



What Is a Prediction Interval?

Definition

A prediction interval gives a range of plausible future outcomes for a single observation, constructed so that realized values fall inside it at approximately the stated confidence rate.

$$\hat{y}_{t+1|t} \pm z_{\alpha/2}\hat{\sigma}_e$$

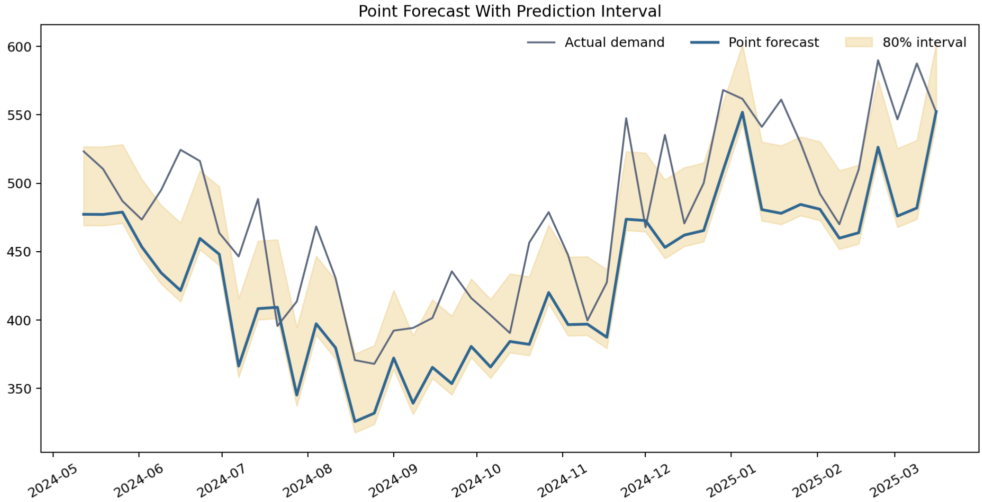
- Point forecast: one best guess (e.g., SP500 return of $+0.2\%$ tomorrow).
- 80% interval: the range that should contain the realized return 80% of the time.
- Managers care about that range because risk limits depend on the downside tail.

Key message: Intervals answer "how wrong could we be?"

Notebook Example: Point Forecast + Intervals (Example 5.1)

- Daily S&P 500 returns over the test period with the random forest point forecast and 80%/95% prediction intervals.
- Days where the realized return falls outside the 80% interval are flagged with red markers.
- The interval widens during high-VIX episodes because market uncertainty rises when volatility spikes.

Notebook Example 5.1: S&P 500 Point Forecast + Intervals



Residual-Based Interval Construction

Definition: Residual uncertainty

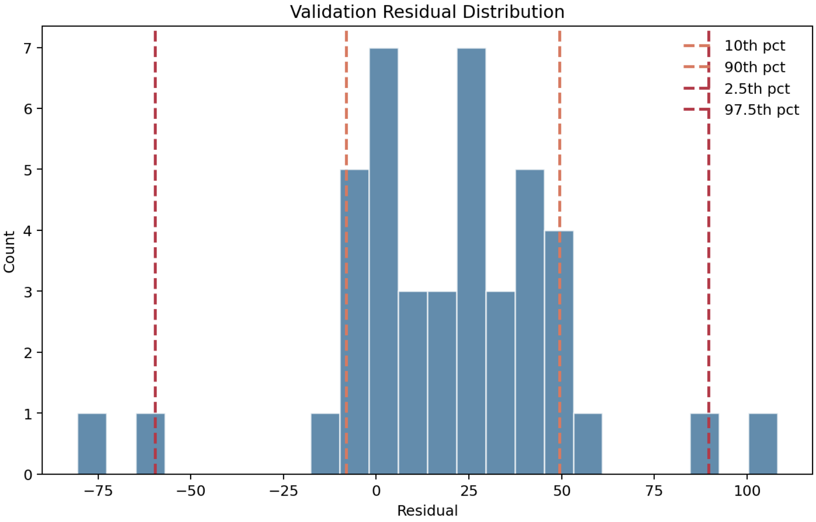
Residual uncertainty is the unpredictable part of the forecast error that remains after the model uses all available information.

$$\text{Lower} = \hat{y}_{t+1|t} + Q_{\alpha/2}(\text{OOS residuals}), \quad \text{Upper} = \hat{y}_{t+1|t} + Q_{1-\alpha/2}(\text{OOS residuals})$$

- We estimate intervals from out-of-sample validation residuals.
- Empirical quantiles are safer than the normal-theory formula for equity returns.
- This keeps calibration consistent with our validation discipline.

Key message: Use empirical quantiles when residuals are fat-tailed—common in financial return forecasting.

Notebook Example: Residuals and Fat Tails (Example 5.2)



Coverage vs Sharpness

Definition: Coverage

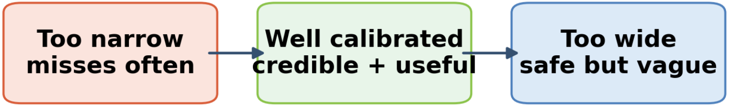
Coverage is the fraction of realized outcomes that fall inside a prediction interval.

Definition: Sharpness

Sharpness refers to how narrow the interval is. Narrower intervals are more informative, but only if coverage stays credible.

Coverage vs Sharpness: Concept

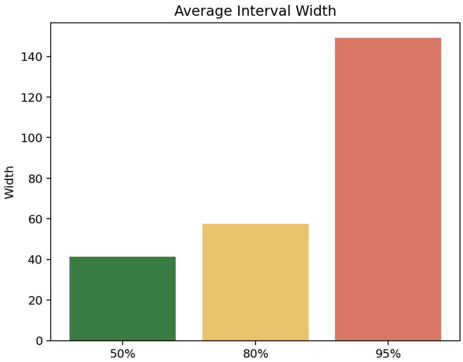
Good intervals balance coverage and sharpness



Goal: hit the advertised rate
without becoming operationally useless

concept adapted from: Medium

Notebook Example: 50% / 80% / 95% Tradeoff (Example 5.3)



- 95% intervals cover more realizations but are too wide for daily position sizing.
- 80% intervals balance coverage and actionable risk information.

State-Dependent Uncertainty

Definition: Heteroskedasticity

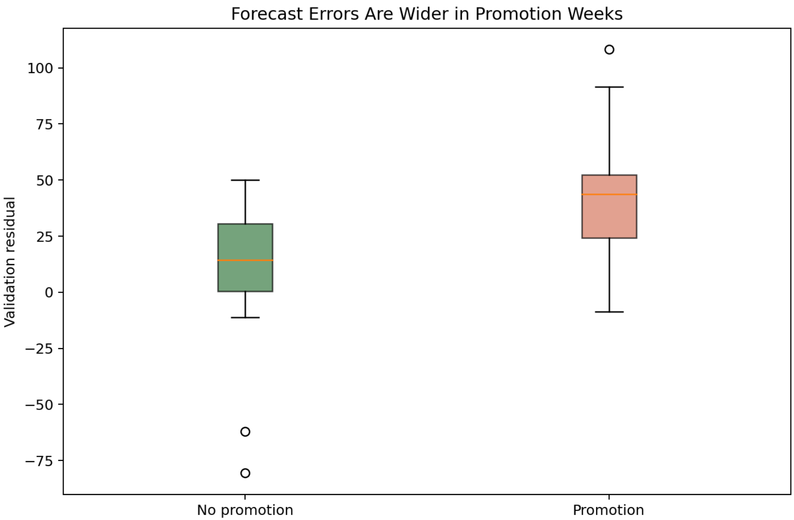
Heteroskedasticity means forecast errors have different variance in different market states, such as calm low-volatility periods versus stressed high-volatility episodes.

- One fixed interval width can hide this state dependence.
- In equity markets, VIX level is a natural regime indicator.
- The same model is “more uncertain” in high-VIX states.

Practical implication

Intervals calibrated only on average residuals understate risk during stress episodes.

Notebook Example: High-VIX Periods Are More Uncertain (Example 5.4)



Scenario Forecasting

Definition

A scenario forecast asks how the predicted path changes when key economic assumptions change, such as VIX level, momentum direction, or cross-asset correlations.

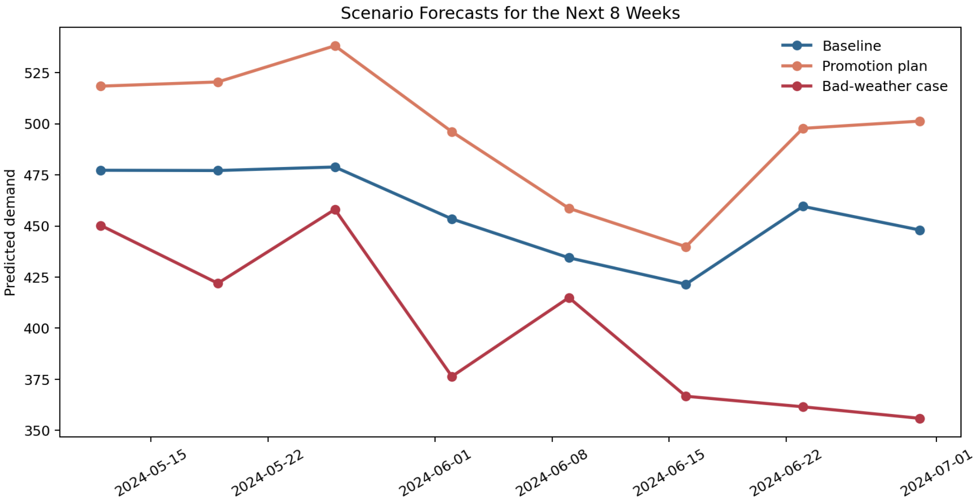
- Not all uncertainty comes from random noise.
- Some uncertainty comes from assumptions we can vary deliberately.
- Scenarios let a risk manager ask: "what if volatility spikes from here?"

Key message: Scenarios are decision tools, not just statistical decoration.

Notebook Example: Bull, Neutral, Bear Scenarios (Example 5.5)

- Three 15-day forward scenarios starting from the last test observation.
- **Bull**: positive MSFT momentum, declining VIX.
- **Neutral**: all features at recent averages.
- **Bear**: negative momentum, rising volatility.
- Cumulative return paths and daily confidence bands are shown for each scenario.

Notebook Example 5.5: Bull / Neutral / Bear Scenarios



Asymmetric Loss: Why Symmetry Can Mislead

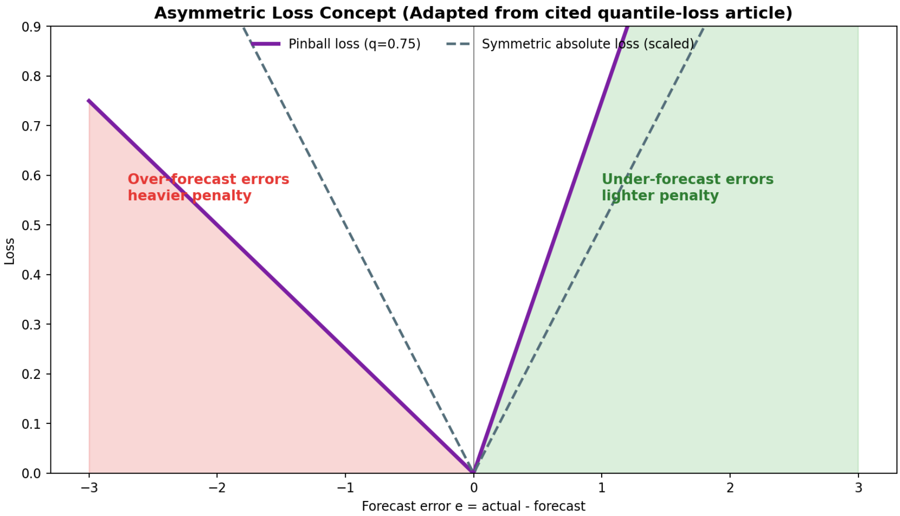
Definition: Asymmetric cost

Asymmetric cost means that over-forecasting and under-forecasting carry different operational penalties. Standard RMSE treats both directions equally, which is often unrealistic.

$$\text{Cost}(e) = \begin{cases} c_{\text{over}} \cdot |e| & e < 0 \quad \text{(over-forecast: held too much, suffered drawdown)} \\ c_{\text{under}} \cdot |e| & e > 0 \quad \text{(under-forecast: held too little, missed rally)} \end{cases}$$

- If $c_{\text{over}} > c_{\text{under}}$: firm prefers caution—drawdowns cost more than missed gains.
- Optimal percentile shifts away from the median toward the conservative direction.

Asymmetric Loss: Concept

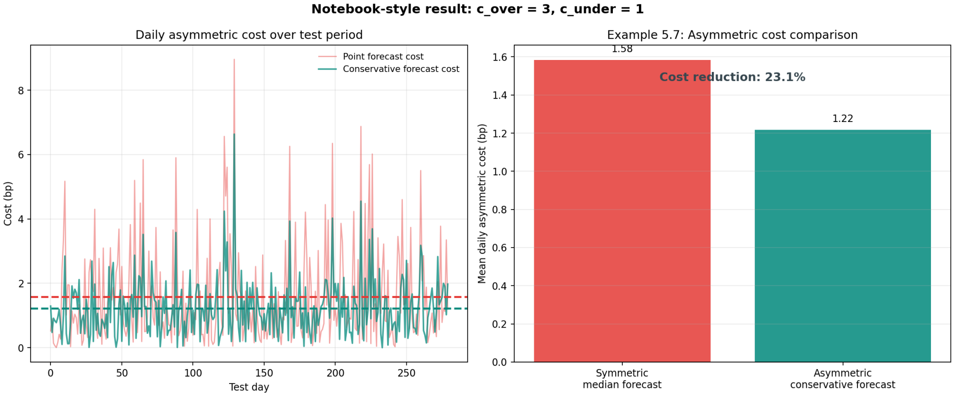


Notebook Example: Asymmetric Cost Framework (Example 5.7)

- Applied to the S&P 500 model with $c_{\text{over}} = 3$, $c_{\text{under}} = 1$.
- Drawdown penalty is three times higher than the opportunity cost of a missed rally.
- The optimal forecast percentile shifts from the median (50th) to the 75th percentile.
- Total asymmetric cost under the conservative forecast is meaningfully lower.

Key message: Minimize asymmetric cost, not symmetric RMSE, when business stakes are unequal.

Notebook Example 5.7: Asymmetric Cost Comparison



From Intervals to Decisions

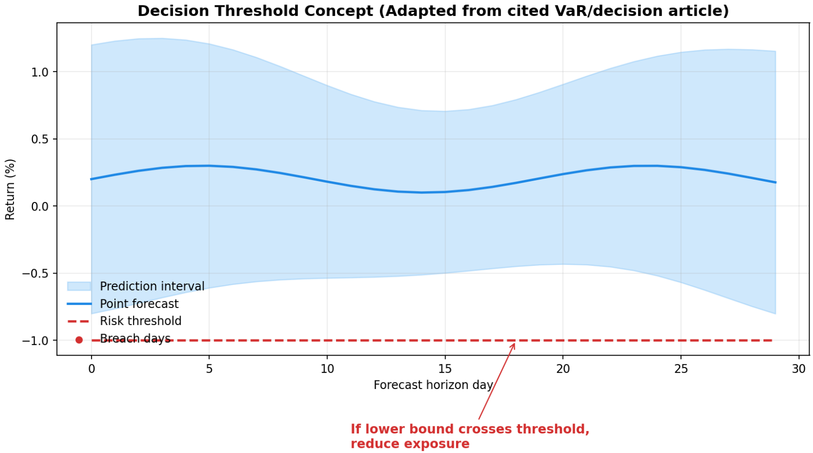
Definition: Decision threshold

A decision threshold is the business cutoff that matters operationally: a drawdown limit, a value-at-risk ceiling, or a minimum return required for action.

- A forecast interval becomes valuable only when it is linked to a decision rule.
- If the lower bound of the return interval breaches the drawdown limit, the manager should reduce exposure—even if the point forecast looks benign.

Key message: The forecast is not the final product. The decision is.

Decision Threshold Concept



concept adapted from: Medium – VaR and decision thresholds

Three-Step Translation Template

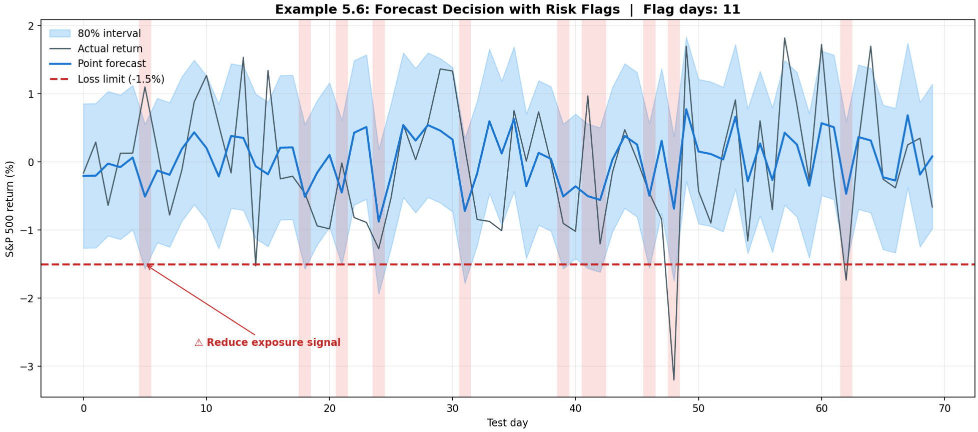
1. **State the forecast.**
   "Model predicts next-day SP500 return of $+0.2\%$, 80% interval: -0.9% to $+1.3\%$."
2. **State the threshold.**
   "Our risk limit is a -1.5% daily drawdown. The 95% lower bound crosses this limit."
3. **State the recommendation.**
   "Reduce overnight leverage to stay within drawdown limits even under the 95th-percentile downside."

Without step 3, the interval is a number with no action value.

Notebook Example: Reduce-Exposure Signals (Example 5.6)

- A loss limit of -1.5% is defined as the exposure threshold.
- For each test-period day, the model checks whether the lower bound of the 80% interval falls below this limit.
- On flag days, a "reduce exposure" signal is issued even if the point forecast is near zero.
- This converts the statistical interval into a binary risk management rule.

Notebook Example 5.6: Forecast Decision with Risk Flags



Takeaways

1. Point forecasts should be paired with empirical uncertainty bands, especially for fat-tailed financial returns.
2. Good intervals balance coverage and sharpness; 80% is often the practical sweet spot.
3. Uncertainty depends on market state: high-VIX periods require wider intervals.
4. Scenario analysis connects forecasting directly to planning assumptions.
5. When costs are asymmetric, shift the decision percentile away from the median.
6. The forecast-to-decision translation step—comparing the interval to a threshold—is what makes forecasting operationally useful.

Key message: Lecture 6 will turn these uncertainty ideas into communication and governance routines.